

Queue Operation

1. Enqueue

```
~n~QUEUE MENU~n~
1. Enqueue
2. Dequeue
3. Peek (Front)
4. Display Queue
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 1
Enter a value to enqueue: 10
[10]
~n~QUEUE MENU~n~
1. Enqueue
2. Dequeue
3. Peek (Front)
4. Display Queue
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 1
Enter a value to enqueue: 20
[10, 20]
~n~QUEUE MENU~n~
1. Enqueue
2. Dequeue
3. Peek (Front)
4. Display Queue
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 1
Enter a value to enqueue: 30
[10, 20, 30]
```

2. Dequeue

```
~n~QUEUE MENU~n~
1. Enqueue
2. Dequeue
3. Peek (Front)
4. Display Queue
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 2
Dequeued value: 10
[20, 30]
```

3. Peek

```
[20, 30]
~n~QUEUE MENU~n~
1. Enqueue
2. Dequeue
3. Peek (Front)
4. Display Queue
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 3
Front element: 20
```

4. Display Queue

```
~n~QUEUE MENU~n~
1. Enqueue
2. Dequeue
3. Peek (Front)
4. Display Queue
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 4
Front -> Rear: [20, 30]
```

Exiting the Program

```
~n~QUEUE MENU~n~
1. Enqueue
2. Dequeue
3. Peek (Front)
4. Display Queue
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 0
~n~MAIN MENU~n~
1. Stack Operations (LIFO)
2. Queue Operations (FIFO)
0. Exit
>>> 0
PS C:\Users\living\OneDrive\文档\codes_101>
```

5 and 6. Check if Empty or Full

```
~n~QUEUE MENU~n~
1. Enqueue
2. Dequeue
3. Peek (Front)
4. Display Queue
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 5
Its not Empty: You have 2-elements in the queue
~n~QUEUE MENU~n~
1. Enqueue
2. Dequeue
3. Peek (Front)
4. Display Queue
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 6
The queue is not full: You have 2-elements in the queue
```

Stack Operation

1. Pop

```
~*~MAIN MENU~*~
1. Stack Operations (LIFO)
2. Queue Operations (FIFO)
0. Exit
>>> 1
~*~STACK MENU~*~
1. Push
2. Pop
3. Peek (Top)
4. Display Stack
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 1
Enter a value to push: 10
[10]
~*~STACK MENU~*~
1. Push
2. Pop
3. Peek (Top)
4. Display Stack
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 1
Enter a value to push: 20
[10, 20]
~*~STACK MENU~*~
1. Push
2. Pop
3. Peek (Top)
4. Display Stack
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 1
Enter a value to push: 30
[10, 20, 30]
```

2. Pop

```
~*~STACK MENU~*~
1. Push
2. Pop
3. Peek (Top)
4. Display Stack
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 2
Popped value: 30
[10, 20]
```

3. Peek

```
~*~STACK MENU~*~
1. Push
2. Pop
3. Peek (Top)
4. Display Stack
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 3
20
```

4. Display Stack

```
~*~STACK MENU~*~
1. Push
2. Pop
3. Peek (Top)
4. Display Stack
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 4
[10, 20]
```

5. Check if Empty

```
~*~STACK MENU~*~
1. Push
2. Pop
3. Peek (Top)
4. Display Stack
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 5
Its not Empty: You have 2-elements in the list
```

6. Check if Full

```
~*~STACK MENU~*~
1. Push
2. Pop
3. Peek (Top)
4. Display Stack
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 6
The list is not full: You have 2-elements in the list
```

0. Back to Main Menu

```
~*~STACK MENU~*~
1. Push
2. Pop
3. Peek (Top)
4. Display Stack
5. Check if Empty
6. Check if Full
0. Back to Main Menu
>>> 0
~*~MAIN MENU~*~
1. Stack Operations (LIFO)
2. Queue Operations (FIFO)
0. Exit
>>> █
```